Moves in the Tic Tac Toe Game

```
| | | |
| | | |
| | | |
Turn: X
Player X took position (2, 0).
| | | |
| | | |
|X| | |
Turn: O
reference:
row 0 is neutral.
row 1 is happy.
row 2 is surprise.
1/1 [==============================] - 0s 57ms/step
Emotion detected as neutral (row 0). Enter 'text' to use text input instead
(0, 1 or 2). Otherwise, press Enter to continue.

reference:
col 0 is neutral.
col 1 is happy.
col 2 is surprise.
1/1 [==============================] - 0s 33ms/step
Emotion detected as happy (col 1). Enter 'text' to use text input instead (0,
1 or 2). Otherwise, press Enter to continue.

Player O took position (0, 1).
| |O| |
| | | |
|X| | |
Turn: X
Player X took position (0, 2).
| |O|X|
| | | |
|X| | |
Turn: O
reference:
row 0 is neutral.
row 1 is happy.
row 2 is surprise.
1/1 [==============================] - 0s 32ms/step
Emotion detected as happy (row 1). Enter 'text' to use text input instead (0,
1 or 2). Otherwise, press Enter to continue.
```

```
reference:
col 0 is neutral.
col 1 is happy.
col 2 is surprise.
1/1 [==============================] - 0s 35ms/step
Emotion detected as happy (col 1). Enter 'text' to use text input instead (0,
1 or 2). Otherwise, press Enter to continue.

Player O took position (1, 1).
| |O|X|
| |O| |
|X| | |
Turn: X
Player X took position (1, 2).
| |O|X|
| |O|X|
|X| | |
Turn: O
reference:
row 0 is neutral.
row 1 is happy.
row 2 is surprise.
WARNING:tensorflow:5 out of the last 5 calls to <function
Model.make_predict_function.<locals>.predict_function at 0x00000188F60FD580>
triggered tf.function retracing. Tracing is expensive and the excessive number
of tracings could be due to (1) creating @tf.function repeatedly in a loop,
(2) passing tensors with different shapes, (3) passing Python objects instead
of tensors. For (1), please define your @tf.function outside of the loop. For
(2), @tf.function has reduce_retracing=True option that can avoid unnecessary
retracing. For (3), please refer to
https://www.tensorflow.org/guide/function#controlling_retracing and
https://www.tensorflow.org/api_docs/python/tf/function for  more details.
1/1 [==============================] - 0s 32ms/step
Emotion detected as happy (row 1). Enter 'text' to use text input instead (0,
1 or 2). Otherwise, press Enter to continue.
2
reference:
col 0 is neutral.
col 1 is happy.
col 2 is surprise.
WARNING:tensorflow:6 out of the last 6 calls to <function
Model.make_predict_function.<locals>.predict_function at 0x00000188FB17E8E0>
```

```
triggered tf.function retracing. Tracing is expensive and the excessive number
of tracings could be due to (1) creating @tf.function repeatedly in a loop,
(2) passing tensors with different shapes, (3) passing Python objects instead
of tensors. For (1), please define your @tf.function outside of the loop. For
(2), @tf.function has reduce_retracing=True option that can avoid unnecessary
retracing. For (3), please refer to
https://www.tensorflow.org/guide/function#controlling_retracing and
https://www.tensorflow.org/api_docs/python/tf/function for  more details.
1/1 [==============================] - 0s 37ms/step
Emotion detected as surprise (col 2). Enter 'text' to use text input instead
(0, 1 or 2). Otherwise, press Enter to continue.
1
Position (1, 2) is already taken.
| |O|X|
| |O|X|
|X| | |
Turn: O
reference:
row 0 is neutral.
row 1 is happy.
row 2 is surprise.
1/1 [==============================] - 0s 32ms/step
Emotion detected as happy (row 1). Enter 'text' to use text input instead (0,
1 or 2). Otherwise, press Enter to continue.
2
reference:
col 0 is neutral.
col 1 is happy.
col 2 is surprise.
1/1 [==============================] - 0s 36ms/step
Emotion detected as happy (col 1). Enter 'text' to use text input instead (0,
1 or 2). Otherwise, press Enter to continue.

Position (1, 1) is already taken.
| |O|X|
| |O|X|
|X| | |
Turn: O
reference:
row 0 is neutral.
row 1 is happy.
row 2 is surprise.
1/1 [==============================] - 0s 32ms/step
```

```
Emotion detected as happy (row 1). Enter 'text' to use text input instead (0, 1 or 2). Otherwise, press Enter to continue.

reference:
col 0 is neutral.
col 1 is happy.
col 2 is surprise.
1/1 [==============================] - 0s 35ms/step
Emotion detected as happy (col 1). Enter 'text' to use text input instead (0, 1 or 2). Otherwise, press Enter to continue.
2
Position (1, 1) is already taken.
| |O|X|
| |O|X|
|X| | |
Turn: O
reference:
row 0 is neutral.
row 1 is happy.
row 2 is surprise.
1/1 [==============================] - 0s 33ms/step
Emotion detected as surprise (row 2). Enter 'text' to use text input instead (0, 1 or 2). Otherwise, press Enter to continue.

reference:
col 0 is neutral.
col 1 is happy.
col 2 is surprise.
1/1 [==============================] - 0s 33ms/step
Emotion detected as happy (col 1). Enter 'text' to use text input instead (0, 1 or 2). Otherwise, press Enter to continue.

Player O took position (2, 1).
| |O|X|
| |O|X|
|X|O| |
Player O has won!
```

Answers To Questions

- How well did your interface work?
    - Not very well, I had to be right up in the camera for it to have any semblance of accuracy.
- Did it recognize your facial expressions with the same accuracy as it achieved against the test set?
    - Well I think it recognized properly about 70% of the time but it really didn't feel like that. Might have felt closer to 60% at some points.
- If not, why not?
    - My theory is that I either need more training data or a better structure to the model. 70% accuracy already isn't really high enough for a model to have super consistent detection from what I've heard. It would be nice if I better knew how to use drop layers to avoid overfitting, or how to layer convolutional layers together better with dense layers.

Get Emotion Function

```
face_id_model =
models.load_model('submissions/Step6TicTacToe/basic_model_100_epochs_timestamp
_1771398105.keras')

resized_img = cv2.resize(img, image_size)
full_img = np.repeat(resized_img[:, :, np.newaxis], 3, axis=2)
full_img = full_img[np.newaxis, ...]
output = face_id_model.predict(full_img)

return int(np.argmax(output))
```